WG05

World Industry, Transport and Environmental Issues

About 1-2 questions a paper. Documented repeats: industrial region to country (RAS 2013, 2016, 2023), greenhouse gases in descending order of contribution (2016, 2018), the acid rain pollutant pair (2016, 2018), biodiversity hotspots (2016, 2018), tropical deforestation (2024).

This is the densest chapter in World Geography and the one RPSC has raided most often. Two things repeat by name: industrial regions matched to countries, and the environment treaties with their years. The environment half also overlaps with the Science and Technology block, so a page learnt here pays twice.

World Industry, Transport & Environment

Europe
- Ruhr - Germany
- Lorraine - France
- Midlands - UK
- Donbas - Ukraine
- Silesia - Poland

Asia & America
- Keihin/Kanto - Japan
- Great Lakes & New England - USA
- Silicon Valley
- Anshan & Shanghai - China

Transport
- Trans-Siberian
- Canadian Pacific
- Trans-Australian
- Suez
- Panama
- Pan-American Highway

Climate regime
- UNFCCC 1992
- Kyoto 1997
- Paris 2015
- COP26-COP31
- IPCC 1988

Ozone & pollution
- Vienna 1985
- Montreal 1987
- Kigali 2016
- Acid rain SO2 + NOx
- Plastics treaty

Land & life
- UNCCD 1994
- Ramsar 1971
- CITES 1973
- CBD 1992
- 36 hotspots

Industrial regions matched to countries – the most repeated item in World Geography

Start here and give it the most time. RPSC has asked industrial-region matching in 2013, 2016 and 2023, sometimes as a match-list and sometimes as 'which pair is NOT correctly matched'. The logic behind every one of these regions is the same - coal, or iron ore, or a port. You do not need the logic to score; you need the pair. Learn the region with its country first, then the leading city.

Industrial region to country - learn this table cold

Industrial region	Country	Basis / leading cities
Ruhr	Germany	Coal-based iron and steel; Essen, Dortmund, Duisburg
Saar	Germany	The other German coalfield
Lorraine	France	'Minette' iron ore, Western Europe's largest ore reserve
Midlands / Black Country	United Kingdom	Birmingham, engineering
Donbas (Donetsk Basin)	Ukraine	Coal and steel
Kuzbas (Kuznetsk Basin)	Russia	Coal; also the Urals with Magnitogorsk
Silesia	Poland	Katowice
Sambre-Meuse	Belgium	Coalfield belt
Po valley	Italy	Northern Italy's manufacturing core
Keihin (Kanto plain)	Japan	Tokyo-Yokohama, Japan's largest belt
Great Lakes / Detroit	USA	Automobiles
New England	USA	Earliest centre of cotton textiles
Silicon Valley	USA	California - electronics and IT
Anshan	China	Manchuria, steel

- United Kingdom in detail: Lancashire (Manchester, cotton - 'Cottonopolis'), Yorkshire (Leeds-Bradford, wool), Clydeside (Glasgow, shipbuilding), Sheffield (cutlery and special steels).
- Japan in detail: Keihin (Tokyo-Yokohama, largest), Hanshin (Osaka-Kobe), Chukyo (Nagoya), Kitakyushu/Yawata (steel).
- USA in detail: Pittsburgh-Lake Erie is the steel belt; New England is textiles; Silicon Valley is electronics.
- China in detail: Shanghai is the largest industrial city; Wuhan and Beijing-Tianjin are the other cores.

Industrial regions grouped by continent

World industrial regions

Western Europe	Eastern Europe	Asia	North America
- Ruhr, Saar - Germany	- Donbas - Ukraine	- Keihin - Japan	- Great Lakes - USA
- Lorraine - France	- Silesia - Poland	- Hanshin - Japan	- New England - USA
- Midlands - UK	- Kuzbas - Russia	- Anshan - China	- Silicon Valley - USA
- Sambre-Meuse - Belgium	- Urals - Russia	- Shanghai - China	- Pittsburgh - USA
- Po valley - Italy			

ASKED IN RAS

This one topic has been asked three times. RAS Pre 2013 - which pair of industrial region and country is NOT correctly matched? The wrong pair was Donbas - Poland (Donbas is Ukraine; Silesia is the Polish one). RAS Pre 2016 and again in 2023 - a match-list of Lorraine, Ruhr, Donbas and Midlands against France, Germany, Ukraine and the United Kingdom. And RAS Pre 2016 asked it plainly: Essen, Dortmund and Duisburg belong to the Ruhr.

TRAP

Three pairs RPSC swaps deliberately. Donbas is Ukraine, not Poland. Kuzbas is Russia, not Ukraine. Lorraine is the iron ORE field of France while the Ruhr is the COAL field of Germany - ore in France, coal in Germany.

Minerals, steel and energy - who leads what

This section is a list, and RPSC asks it as a list. Keep to the leaders that have been stable for years. Notice one pattern: the biggest producer of a mineral is often not the biggest exporter of it, because a large economy consumes its own output.

Mineral and energy leaders

Item	Leader	Note
Crude steel	China	More than half of world output; India No. 2, Japan No. 3
Iron ore	Australia	Largest producer and exporter; Brazil No. 2 (Carajas, Itabira)
Copper	Chile	Chuquicamata and Escondida; also largest reserves
Uranium	Kazakhstan	-
Coal and gold	China	-
Nickel	Indonesia	-
Bauxite reserves	Guinea	Largest reserves in the world
Crude oil	USA	Largest producer; Saudi Arabia is the largest exporter

- OPEC was founded at Baghdad in 1960; its headquarters are at Vienna.
- Steel needs coal plus iron ore plus a cheap route between them - that is why every classic steel region sits on a coalfield or a waterway.
- Duisburg on the Rhine is the world's great inland port and is what makes the Ruhr work.

Transport - railways, the great road, canals, ports

Trans-continental railways were built to bind a large country together, so each one runs coast to coast or capital to far edge. Three names come up. Learn each with its country and its two terminal cities - RPSC's format is 'which pair is NOT correctly matched'.

Trans-continental railways and the longest road

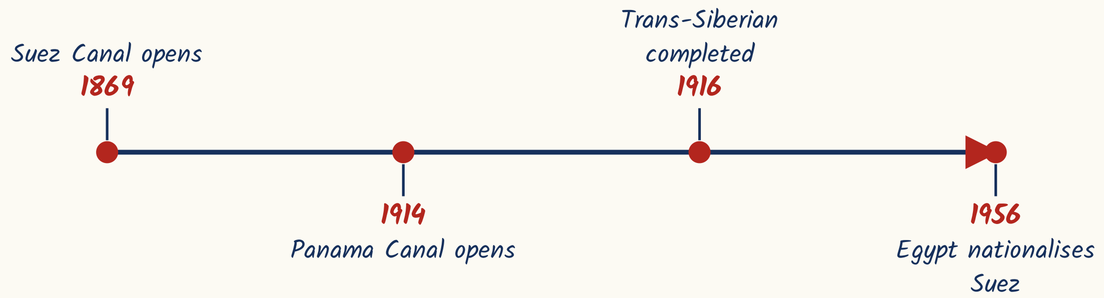
Line	Country	Route and fact
Trans-Siberian Railway	Russia	Moscow to Vladivostok, about 9,289 km - the longest railway line in the world; completed 1916; crosses eight time zones
Trans-Australian (Indian Pacific)	Australia	Perth to Sydney; the world's longest dead-straight stretch of track, about 478 km across the Nullarbor Plain
Canadian Pacific Railway	Canada	Montreal to Vancouver
Pan-American Highway	Alaska to southern Argentina	The longest motorable road; broken only by the Darien Gap

Suez vs Panama - the classic two-canal question

Suez Canal	Panama Canal
Opened 1869	Opened 1914
About 193 km	About 82 km
Port Said to Suez	Atlantic to Pacific
Links the Mediterranean with the Red Sea	Links the Atlantic with the Pacific
Sea-level, NO locks	Has locks - Gatun, Pedro Miguel, Miraflores, plus Gatun Lake
Nationalised by Egypt in 1956	-

- Kiel Canal joins the North Sea to the Baltic.
- The Grand Canal of China is the world's longest canal.
- Shanghai is the world's busiest container port; Singapore second; Rotterdam the largest in Europe.
- Hartsfield-Jackson Atlanta is the busiest airport by total passengers.
- Dubai is the busiest by international passengers; Hong Kong is the busiest for air cargo.

The great canals and railways in date order



YAAD RAKHO

Suez has No locks because it is at Sea level - S with S. Panama has locks because it must lift ships over the isthmus into Gatun Lake.

Global warming and the climate regime – UNFCCC to Paris to the COPs

Understand the architecture before the dates and the whole thing becomes easy. The UNFCCC of 1992 is the parent convention – it only set the goal and the principle. Under it sit two operative agreements: the Kyoto Protocol of 1997, which put binding cuts on developed countries only, and the Paris Agreement of 2015, under which every country sets its own target. The COP is simply the annual meeting of the parties to the UNFCCC, numbered in sequence.

The climate regime in order

Stockholm Conference 1972 – UNEP created →

IPCC set up 1988 by WMO and UNEP →

UNFCCC adopted at Rio 1992, in force 1994 →

Kyoto Protocol COP3, 1997, in force 2005 →

Paris Agreement COP21, 2015, in force 2016 →

Global Stocktake begins, COP28 Dubai 2023

- Stockholm Conference on the Human Environment 1972 created UNEP, headquarters Nairobi, and gave us World Environment Day on 5 June.
- UNFCCC rests on the principle of common but differentiated responsibilities.
- IPCC was set up in 1988 by the World Meteorological Organization and UNEP; it shared the 2007 Nobel Peace Prize with Al Gore; the AR6 Synthesis Report appeared in 2023.
- Kyoto Protocol - adopted at COP3 in 1997, in force 16 February 2005, first commitment period 2008-12.
- Kyoto's three flexible mechanisms: Clean Development Mechanism, Joint Implementation, Emissions Trading.
- Doha Amendment 2012 extended Kyoto to 2020.
- Paris Agreement - adopted at COP21 on 12 December 2015, in force 4 November 2016.
- Paris goal: hold warming well below 2 degrees Celsius and pursue 1.5 degrees, through Nationally Determined Contributions and a global stocktake every five years.
- Kyoto basket of gases: carbon dioxide, methane, nitrous oxide, HFCs, PFCs and sulphur hexafluoride; nitrogen trifluoride was added by the Doha Amendment.
- Sulphur hexafluoride has the highest global warming potential of the basket.
- CFCs are NOT in the Kyoto basket - they are handled by the Montreal Protocol.

COP sequence – hosts and headline outcome

COP21 Paris 2015 – Paris Agreement → COP26 Glasgow 2021 – Glasgow Climate Pact, 'phase-down' of unabated coal; India's net-zero 2070 and Panchamrit → COP27 Sharm el-Sheikh 2022 – Loss and Damage Fund created → COP28 Dubai 2023 – first Global Stocktake, 'transitioning away from fossil fuels', treble renewables → COP29 Baku 2024 – New Collective Quantified Goal of at least 300 billion dollars a year by 2035; Article 6 carbon market rules finalised → COP30 Belem 2025 – the 'Global Mutirao' decision, adaptation finance to be trebled by 2035, no fossil-fuel roadmap

Net-zero target years and India's NDC

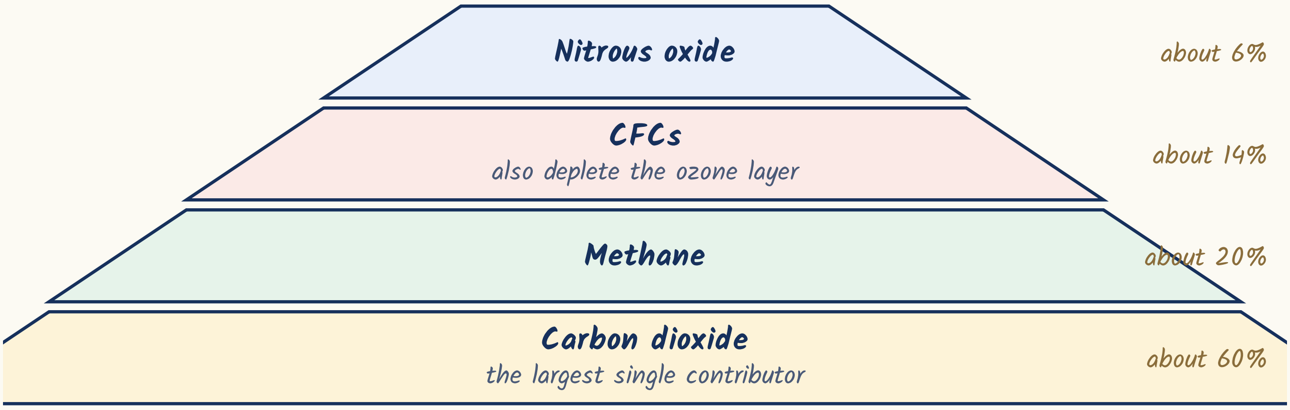
Country / bloc	Net-zero year
India	2070
China	2060
European Union, United Kingdom, United States, Japan	2050

- India's updated NDC – cut the emission intensity of GDP by 45 per cent by 2030 over 2005 levels.
- India's updated NDC – reach 50 per cent non-fossil installed power capacity by 2030.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2018 - arrange the greenhouse gases in decreasing order of their conventionally estimated contribution to the enhanced greenhouse effect. The order is carbon dioxide (about 60 per cent), then methane (about 20 per cent), then CFCs (about 14 per cent), then nitrous oxide (about 6 per cent). Also remember nitrogen is NOT a greenhouse gas.

Contribution to the enhanced greenhouse effect



CURRENT LINK

Current link, as of September 2026: COP31 will be held at Antalya, Turkiye, from 9 to 20 November 2026 - after the exam-season news cycle begins. Turkiye hosts and its minister Murat Kurum is COP President, while Australia's Chris Bowen presides over the negotiations under a shared arrangement agreed at COP30. A pre-COP is planned in Fiji in early October 2026. Track it: the host city plus the headline decision is exactly the form RPSC uses.

Ozone, acid rain and pollution

Keep ozone and climate change in separate boxes in your head - RPSC exploits the overlap. Ozone depletion is a stratospheric problem caused by chlorine and bromine compounds, handled by Vienna and Montreal. Global warming is a tropospheric heat problem caused by greenhouse gases, handled by UNFCCC, Kyoto and Paris. The Kigali Amendment is the one bridge between them, because HFCs are greenhouse gases dealt with under an ozone treaty.

The ozone regime

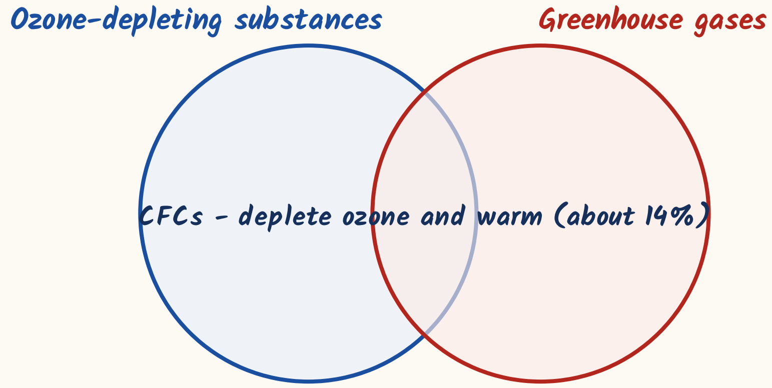
Vienna Convention 1985 - the framework → Montreal Protocol adopted 16 September 1987 →

In force 1 January 1989 → First UN treaty with universal ratification →

Kigali Amendment 2016 - HFC phase-down → Kigali in force 1 January 2019; India ratified 2021

- Ozone lies in the stratosphere and is measured in Dobson Units.
- World Ozone Day, the International Day for the Preservation of the Ozone Layer, is 16 September.
- Kigali was adopted at Kigali, Rwanda, and phases down hydrofluorocarbons - greenhouse gases that do NOT deplete ozone.
- Carbon dioxide is not an ozone-depleting substance; CFCs, halons and carbon tetrachloride are.

Ozone-depleting substances and greenhouse gases



etrachloride deplete ozone only; carbon dioxide and methane only warm; HFCs do not deplete ozone yet Kigal

- Acid rain – sulphur dioxide and oxides of nitrogen are the precursors; they form sulphuric and nitric acid.
- Rain counts as acid rain when its pH falls below 5.6.
- Damage: the Taj Mahal ('marble cancer'), the Black Forest in Germany, Scandinavian lakes.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2018 – statements on acid rain. Sulphur dioxide and oxides of nitrogen are the principal precursors, and that statement is correct. The statement blaming carbon dioxide is the false one. This pollutant pair is one of the most repeated single facts in the whole environment syllabus.

CURRENT LINK

Current link, as of September 2026: there is still no global plastics treaty. UNEA-5.2 at Nairobi adopted resolution 5/14 'End Plastic Pollution' in March 2022, but the negotiating sessions at Busan in 2024 and Geneva in 2025 both ended without agreement, chiefly on whether to cap plastic production. A short session at Geneva in February 2026 (INC-5.3) only elected a new chair, Julio Cordano of Chile, with substantive talks pushed to a further session later in 2026. Also remember the Great Pacific Garbage Patch lies in the North Pacific gyre between Hawaii and California.

Land, biodiversity and the treaty table

The last section is one table you must be able to write from memory. RPSC asks treaties in two forms - convention matched to year, and convention matched to subject. So learn each treaty as a triplet: year, name, subject. Go down the table in chronological order and it will stick.

Environmental conventions - year, name, subject

Year	Treaty	Subject
1971	Ramsar Convention (Ramsar, Iran)	Wetlands; in force 1975; World Wetlands Day 2 February
1973	CITES (Washington)	Trade in endangered species; in force 1975
1985	Vienna Convention	Framework for protection of the ozone layer
1987	Montreal Protocol	Phase-out of ozone-depleting substances
1989	Basel Convention	Trans-boundary movement of hazardous wastes
1992	CBD (Rio)	Biological diversity
1992	UNFCCC (Rio)	Climate change; in force 1994
1994	UNCCD (Paris, 17 June)	Desertification; in force 1996
1997	Kyoto Protocol	Binding emission cuts for developed countries

Year	Treaty	Subject
1998	Rotterdam Convention	Prior informed consent for hazardous chemicals
2001	Stockholm Convention	Persistent organic pollutants
2013	Minamata Convention	Mercury
2015	Paris Agreement	Climate change; NDCs and the 1.5 degree goal
2016	Kigali Amendment	Phase-down of HFCs

Conventions grouped by what they protect

Environmental conventions

Nature and species
- Ramsar 1971 - wetlands
- CITES 1973 - trade
- CBD 1992 - biodiversity
- UNCCD 1994 - drylands

Ozone layer
- Vienna 1985 - framework
- Montreal 1987 - ODS
- Kigali 2016 - HFCs

Climate
- UNFCCC 1992 - Rio
- Kyoto 1997 - cuts
- Paris 2015 - NDCs

Chemicals and waste
- Basel 1989 - wastes
- Rotterdam 1998 - PIC
- Stockholm 2001 - POPs
- Minamata 2013 - mercury

- Under the CBD: Cartagena Protocol on Biosafety 2000, Nagoya Protocol on access and benefit sharing 2010.
- Kunming-Montreal Global Biodiversity Framework - CBD COP15, Montreal 2022, 23 targets including '30x30'.
- World Day to Combat Desertification and Drought is 17 June; UNCCD COP16 met at Riyadh in 2024.
- Land Degradation Neutrality is SDG target 15.3.
- Great Green Wall - launched by the African Union in 2007, an 8,000 km belt of restored land across the Sahel from Senegal to Djibouti.
- Biodiversity hotspots - the concept is Norman Myers's, 1988; there are 36 at present.
- Criteria: at least 1,500 endemic vascular plant species, and loss of at least 70 per cent of the original vegetation.
- India has four - Himalaya, Indo-Burma, Western Ghats and Sri Lanka, and Sundaland (Nicobar Islands).
- Tropical forests record the highest rate of deforestation of all forest types.

ASKED IN RAS

Asked in RAS Pre 2016 and 2018 - the biodiversity hotspot statements: Norman Myers's concept, the 1,500 endemic plant species threshold, and the 70 per cent vegetation loss criterion were all correct. And in RAS Pre 2024 - tropical forests record the highest deforestation rate among all forest types. Both are worth memorising word for word.

YAAD RAKHO

Two anchors will place most treaties for you. Everything ozone is in the 1980s - Vienna 1985, Montreal 1987. Everything Rio is 1992 - CBD and UNFCCC together. Ramsar 1971 and CITES 1973 sit before them; the chemicals trio Basel-Rotterdam-Stockholm runs 1989-1998-2001.

REVISION RECAP

1. Ruhr-Germany (Essen, Dortmund, Duisburg), Lorraine-France (minette ore), Midlands-UK, Donbas-Ukraine, Kuzbas-Russia, Silesia-Poland.
2. Keihin/Kanto is Japan's largest belt; New England is US textiles, Great Lakes/Detroit is autos, Silicon Valley is electronics.
3. China makes over half the world's crude steel; Australia leads iron ore; Chile copper; Kazakhstan uranium; Guinea bauxite reserves.
4. USA is the largest crude oil producer, Saudi Arabia the largest exporter. OPEC - Baghdad 1960, headquarters Vienna.
5. Trans-Siberian: Moscow-Vladivostok, about 9,289 km, longest in the world, eight time zones.
6. Trans-Australian has the longest straight track (Nullarbor); Canadian Pacific is Montreal-Vancouver; Pan-American Highway breaks at the Darien Gap.
7. Suez 1869, 193 km, no locks. Panama 1914, 82 km, has locks. Kiel joins North Sea to Baltic; Grand Canal of China is the longest canal.
8. Shanghai busiest container port; Atlanta busiest airport by total passengers, Dubai by international, Hong Kong by cargo.
9. UNFCCC 1992 is the parent; Kyoto 1997 binds only developed countries; Paris 2015 runs on NDCs.
10. COP26 Glasgow - India net-zero 2070. COP27 - Loss and Damage Fund. COP28 - first Global Stocktake. COP29 - 300 billion dollars by 2035. COP30 Belem 2025. COP31 Antalya, November 2026.
11. Greenhouse contribution order: CO2 about 60 per cent > CH4 about 20 > CFCs about 14 > N2O about 6. Nitrogen is not a greenhouse gas.
12. Ozone: Vienna 1985, Montreal 1987 (in force 1989, universal ratification), Kigali 2016 for HFCs. World Ozone Day 16 September, Dobson Units, stratosphere.
13. Acid rain = SO2 + NOx, pH below 5.6, marble cancer on the Taj.
14. Treaty years: Ramsar 1971, CITES 1973, Basel 1989, CBD and UNFCCC 1992, UNCCD 1994, Rotterdam 1998, Stockholm 2001, Minamata 2013.
15. Hotspots: Myers 1988, 36 today, 1,500 endemic plants plus 70 per cent vegetation loss; India has four. Tropical forests lose the most cover.